

# EPOD GUIDELINES

## Revision History

| Date     | Version | Description                                                                                 | Author            |
|----------|---------|---------------------------------------------------------------------------------------------|-------------------|
| 07/22/13 | 1.0     | Initial Release                                                                             | Dmitry Novikov    |
| 08/15/13 | 1.1     | EPOD Web Service section completed, field names changed, new field STI added                | Dmitry Novikov    |
| 10/01/13 | 1.2     | JSON formatted message corrected                                                            | Dmitry Novikov    |
| 11/29/13 | 1.3     | New Inspection Type field added, only web service communication option allowed              | Dmitry Novikov    |
| 01/08/14 | 1.4     | Added HTTP POST header example                                                              | Matthew Caldwell  |
| 02/28/14 | 1.5     | XML formatted message corrected                                                             | Matthew Caldwell  |
| 08/25/14 | 1.6     | Update element/field information                                                            | Matthew Caldwell  |
| 12/08/14 | 1.7     | JSON formatted message corrected                                                            | Matthew Caldwell  |
| 06/23/15 | 1.8     | HTTP response status codes                                                                  | Mireille Elhaddad |
| 06/24/15 | 1.9     | Inspection location codes added in Appendix                                                 | Mireille Elhaddad |
| 11/17/15 | 1.10    | Error codes                                                                                 | Mireille Elhaddad |
| 02/05/16 | 1.11    | Updated support for new inspection types                                                    | Matthew Caldwell  |
| 12/01/16 | 1.12    | Updated definition for delivery receipt                                                     | Mireille Elhaddad |
| 08/04/17 | 1.13    | Proof of successful transmission requirement added                                          | Mireille Elhaddad |
| 01/27/19 | 1.14    | Shipment order number added                                                                 | Dmitry Novikov    |
| 05/18/20 | 1.15    | Introduction modified to mention vehicle inspections                                        | Dmitry Novikov    |
| 06/16/21 | 1.16    | Corrected departure date requirements                                                       | Dmitry Novikov    |
| 09/23/21 | 1.17    | Vehicle delivery types and their requirements added, better explanation for web service API | Dmitry Novikov    |



| Date    | Version | Description                                                             | Author         |
|---------|---------|-------------------------------------------------------------------------|----------------|
| 7/22/22 | 1.18    | Added damage classification field, truck number renamed to equipment id | Dmitry Novikov |



## Table of Contents

|                                                   |    |
|---------------------------------------------------|----|
| Revision History.....                             | 1  |
| Table of Contents.....                            | 3  |
| Introduction.....                                 | 4  |
| Vehicle Delivery Types.....                       | 5  |
| Final Delivery Requirements.....                  | 5  |
| Non-Final Delivery Requirements.....              | 5  |
| EPOD Delivery Message.....                        | 6  |
| Data Transmission.....                            | 8  |
| Communication Options.....                        | 8  |
| EPOD Web Service.....                             | 9  |
| Overview.....                                     | 9  |
| Interface.....                                    | 9  |
| Post Delivery Message.....                        | 9  |
| Get Delivery Message.....                         | 11 |
| Demo Client.....                                  | 12 |
| Appendix.....                                     | 13 |
| Inspection Type Codes.....                        | 13 |
| JSON Formatted EPOD Delivery Message Example..... | 15 |
| XML Formatted EPOD Delivery Message Example.....  | 17 |



## Introduction

This document contains implementation guidelines for the interchange of electronic data transmissions from EPOD data providers (DP) to Fenkell Automotive Services (FAS) related specifically to:

1. vehicle electronic proof of delivery (EPOD) information.
2. vehicle inspection information.

Data input is required with a minimal delay, preferably in real-time.

EDI support can be reached at [epod-support@fenkell.com](mailto:epod-support@fenkell.com)

**Note:** The latest version of this document can be obtained directly from Fenkell Dev Support website at <https://dev.fenkell.com/projects/vt/wiki/Interface/EPOD-FAS> (scroll down to Attachments section).



## Vehicle Delivery Types

From EPOD prospective there are two types of vehicle delivery:

- final delivery (to a dealer or other final destination designated by OEM)
- non-final delivery

Each delivery type has its own requirements for the values provided in EPOD Delivery Message.

### Final Delivery Requirements

1. Origin Location should contain location code assigned by OEM. In most of the cases it is a 9-digit [SPLC](#) code but it also could be a plant code or a dealer/upfitter code.
2. Destination Location should contain dealer's code assigned by OEM.
3. Inspection type should have value 05 (Dealer Inspection).
4. Signed delivery receipt should be provided.

### Non-Final Delivery Requirements

1. Origin Location should contain location code assigned by OEM. In most of the cases it is a 9-digit [SPLC](#) code but it also could be a plant code or a dealer/upfitter code.
2. Destination Location should contain location code assigned by OEM. In most of the cases it is a 9-digit [SPLC](#) code but it also could be a plant code or an upfitter code.
3. Inspection type depends on the phase of delivery:
  - Origin Inspections (inspection types 01, 06)
  - Intermediate Inspections (inspection types 02,07,7R)
  - Destination Inspections (inspections types 04,4E,4V,4R,08)



## EPOD Delivery Message

The following table provides list of supported EPOD Delivery Message elements (all elements are required, however “Y” indicates that a value is required for that element.):

| Group                                           | Element             | Description                               | Required        | Example                     |
|-------------------------------------------------|---------------------|-------------------------------------------|-----------------|-----------------------------|
| Main                                            |                     |                                           |                 |                             |
|                                                 | Referenceld         | Key/ID from data sender database          | Y               |                             |
|                                                 | Comment             | Dealer comment (string up to 500 chars)   | N               |                             |
|                                                 | CarrierComment      | Carrier comment (string up to 500 chars)  | N               |                             |
|                                                 | InspectionType      | Inspection type code                      | Y               | (See Inspection Type Codes) |
|                                                 | SubjectToInspection | Subject To Inspection by dealer           | Y <sup>1</sup>  | true or false               |
| Carrier                                         |                     |                                           |                 |                             |
|                                                 | CarrierCode         | SCAC or code assigned by OEM              | Y               | BNSF                        |
|                                                 | EquipmentId         | Truck / railcar number, vessel name       | N               |                             |
|                                                 | TrailerNumber       |                                           | N               |                             |
|                                                 | DriverName          |                                           | N               |                             |
| Shipment                                        |                     |                                           |                 |                             |
|                                                 | OrderNumber         | Shipment order number                     | N               | 679807                      |
|                                                 | OriginCode          | Origin location code assigned by OEM      | Y <sup>5</sup>  | 6B                          |
|                                                 | DestinationCode     | Destination location code assigned by OEM | Y <sup>2,</sup> | 56789                       |
|                                                 | DepartureDateTime   | Departure date and time (ISO 8601)        | Y <sup>5</sup>  | 2014-07-28T06:07:34+04:00   |
|                                                 | DeliveryDateTime    | Delivery date and time (ISO 8601)         | Y <sup>2</sup>  | 2014-07-28T13:54:22+04:00   |
|                                                 | SpecialInstructions | Special delivery instructions             | N               |                             |
| Delivery Receipt (PDF form of delivery receipt) |                     |                                           | Y <sup>3</sup>  |                             |
|                                                 | Referenceld         | Key/ID from data sender database          | Y               |                             |
|                                                 | URL                 | PDF hosted on data sender server          | Y               |                             |
|                                                 | Title               | Filename or Description                   | N               |                             |
| Vehicle (multiple records allowed)              |                     |                                           | Y               |                             |

| Group                                         | Element            | Description                          | Required       | Example |
|-----------------------------------------------|--------------------|--------------------------------------|----------------|---------|
|                                               | VIN                | Must be valid VIN17 (checksum)       | Y              |         |
| Damage (multiple records allowed per vehicle) |                    |                                      | Y <sup>4</sup> |         |
|                                               | AreaCode           | Damage area code (length=2 chars)    | Y              | 01      |
|                                               | TypeCode           | Damage type code (length=2 chars)    | Y              | 02      |
|                                               | SeverityCode       | Damage severity code (length=1 char) | Y              | 3       |
|                                               | ClassificationCode | Damage classification code           | N              | 3       |
|                                               | Comment            | Damage comment                       | N              |         |
| Photo (multiple photos allowed per vehicle)   |                    |                                      | Y <sup>4</sup> |         |
|                                               | ReferenceId        | Filename or Description              | Y              |         |
|                                               | URL                | Image hosted on data sender server   | Y              |         |
|                                               | Title              | Filename or Description              | N              |         |

1. Subject To Inspection (STI) element only applies to Dealer Inspections (inspection type 05)
2. Optional for Intermediate Inspections (inspection types 02,07,7R)
3. Delivery Receipt is required for Dealer Inspections (inspection type 05)
4. Required for damaged vehicles
5. Optional for Destination Inspections (inspections types 04,4E,4V,4R,08)



## **Data Transmission**

### **Communication Options**

Currently FAS supports the following options for data transmission:

1. EPOD Web Service – best option to minimize delays.





## EPOD Web Service

### Overview

EPOD Web Service is built on [JAX-RS](#) stack and conforms to Representational State Transfer ([REST](#)) architectural pattern. The following formats of data exchange are supported:

- JSON
- XML

### Interface

The following functions are currently supported:

| Function              | Access URL                                                                                                            | HTTP Method |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------|-------------|
| Post Delivery Message | <a href="https://wwwtest.fenkell.com/epod/rest/delivery/post">https://wwwtest.fenkell.com/epod/rest/delivery/post</a> | POST        |
| Get Delivery Message  | <a href="https://wwwtest.fenkell.com/epod/rest/delivery/get">https://wwwtest.fenkell.com/epod/rest/delivery/get</a>   | GET         |

**Note:** Given access URLs are for development and testing purposes only. Production URLs will be provided upon completion of testing. Contact [epod-support@fenkell.com](mailto:epod-support@fenkell.com) to obtain credentials.

### Post Delivery Message

Use this function to submit EPOD delivery message about vehicle delivery event or vehicle inspection.

#### HTTP request headers:

Content-Type: application/json *or* application/xml

Accept: application/json *or* application/xml

#### URL parameters:

| Name     | Description   | Type   |
|----------|---------------|--------|
| user     | User login id | String |
| password | User password | String |

#### Parameters encoded in HTTP request body:

EPOD delivery message, could be provided in:

- JSON format (see JSON Formatted EPOD Delivery Message Example in Appendix)
- XML format (see XML Formatted EPOD Delivery Message Example in Appendix)

#### Returns:

If EPOD Delivery message was posted successfully web service responds with HTTP status code 200 and provides EPOD transmission ID.

#### Important:



1. Keep obtained ID as a proof of successful transmission.
2. If an error will be encountered during EPOD message processing, HTTP status code other than 200 will reported back to caller along with a [reason phrase](#) containing detailed error description.

The following is a list of HTTP status codes and reason phrases used to report errors:

| HTTP Status Code | Error Type   | Reason Phrase                              |
|------------------|--------------|--------------------------------------------|
| 401              | Client error | user is not authenticated                  |
| 401              | Client error | user has no associated vendor              |
| 403              | Client error | unauthorized use of carrier code           |
| 415              | Client error | cannot consume content type                |
| 422              | Client error | carrier code is required                   |
| 422              | Client error | unknown carrier code                       |
| 422              | Client error | inspection type is required                |
| 422              | Client error | unknown inspection type                    |
| 422              | Client error | shipment order number is required          |
| 422              | Client error | unknown shipment order number              |
| 422              | Client error | origin location code is required           |
| 422              | Client error | unknown origin location code               |
| 422              | Client error | destination location code is required      |
| 422              | Client error | unknown destination location code          |
| 422              | Client error | departure date/time is required            |
| 422              | Client error | departure date/time can't be in the future |
| 422              | Client error | departure date/time is too old             |
| 422              | Client error | delivery date/time is required             |
| 422              | Client error | delivery can't precede departure           |
| 422              | Client error | delivery date/time can't be in the future  |
| 422              | Client error | delivery date/time is too old              |
| 422              | Client error | delivery receipt is required               |
| 422              | Client error | vehicle record is required                 |
| 422              | Client error | invalid VIN                                |
| 422              | Client error | duplicate VIN reported                     |
| 422              | Client error | invalid damage area code                   |
| 422              | Client error | unknown damage area code                   |
| 422              | Client error | invalid damage type code                   |
| 422              | Client error | unknown damage type code                   |
| 422              | Client error | invalid damage severity code               |
| 422              | Client error | unknown damage severity code               |



| HTTP Status Code | Error Type   | Reason Phrase                     |
|------------------|--------------|-----------------------------------|
| 422              | Client error | no damage photos provided for VIN |
| 429              | Client error | too many requests                 |
| 500              | Server error | <server error description>        |
| 503              | Server error | Service temporary unavailable     |

### Get Delivery Message

Use this function to retrieve EPOD message submitted earlier.

#### HTTP request headers:

Accept: application/json *or* application/xml

#### URL parameters:

| Name        | Description                   | Type   |
|-------------|-------------------------------|--------|
| id          | Delivery message id           | String |
| referenceId | Delivery message reference id | String |
| user        | User login id                 | String |
| password    | User password                 | String |

Either **id** or **referenceId** has to be provided.

#### Returns:

Returns delivery message in JSON or XML format depending on “Accept” header value in client HTTP request. Otherwise, returns an error response.



## Demo Client

Demo Client is a Java console application developed specifically to demonstrate usage of EPOD Web Service interface and could be downloaded from Fenkell Dev Support website at:

<https://dev.fenkell.com/projects/vt/wiki/Interface/EPOD-FAS> (scroll down to Attachments section).

## Appendix

### Inspection Type Codes

Code defining the type of location where an inspection is performed.

| Code | Description                                  | Definition                                                                                                                                                                                                    |
|------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Origin Inspection                            | Location where motor vehicle is inspected prior to loading                                                                                                                                                    |
| 2    | Intermediate Interchange Inspection          | Location in transit between point of origin and destination*                                                                                                                                                  |
| 3    | Railroad Interchange / Marine Survey Preload | Point at which multi-level is transferred from one railroad to another/ Last point of rest prior to loading onto a vessel for ocean transportation                                                            |
| 4    | Destination Ramp                             | Location where motor vehicle is to be unloaded from multi-level/ (4R) In bay or destination on ground/ (4E) Data Entry truckers load sheets/ (4V) verification inspection with truckers**                     |
| 4E   |                                              | Data entry – haulaway load sheets**                                                                                                                                                                           |
| 4R   |                                              | In bay or at destination on ground**                                                                                                                                                                          |
| 4V   |                                              | Verification inspections with haulaway drivers**                                                                                                                                                              |
| 5    | Dealer Inspection                            | Location where carrier transfers possession to manufacturers selling agent                                                                                                                                    |
| 6    | Factory Gate                                 | Location where motor vehicle is considered to be transferred to first carrier / Inspection prior to acceptance by pre delivery processor / (6F) Plant Inspection / (6Y) Yard inspection outside or near Plant |
| 7    | Origin On Rail                               | Performed on multi-level after loading and securement of motor vehicle / (7R) Origin on Rail*                                                                                                                 |
| 7R   |                                              | Origin On Rail*                                                                                                                                                                                               |
| 8    | Destination On Rail                          | Performed on multi-level at destination prior to unloading of motor vehicle**                                                                                                                                 |
| 9    | Marine Survey Discharge                      | First point of rest after discharge from ocean vessel / Major Damage Repair / (9Y) Inside Yard Inspection after repair                                                                                        |
| 11   | Major Damage                                 | Code used by some haul-away carriers to denote presence of major damage to vehicle                                                                                                                            |
| 21   | Major Damage Inspection                      | Code used by some carriers to indicate major damage and additional reporting available                                                                                                                        |
| 51   | Origin Non Distribute                        | Code used by some manufacturers to indicate vehicle hold at origin                                                                                                                                            |
| 52   | Interchange Non Distribute                   | Code used by some manufacturers to indicate vehicle hold at interchange                                                                                                                                       |



| Code | Description                | Definition                                                                                                                                 |
|------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 90   | Delivery With Notification | Code used to note additional information available upon dealer delivery                                                                    |
| 96   | Intermediate Delivery      | Code used for vehicle storage yard arrival                                                                                                 |
| 96Y  | Inbound Yard Inspection    | Code used for vehicle storage yard entry inspection                                                                                        |
| 97   | Outbound Intermediate      | Code used for vehicle storage yard exit                                                                                                    |
| 97Y  | Outbound Yard Inspection   | Code used for vehicle storage yard exit                                                                                                    |
| 98   | GM Dealer Receipt          | Location where carrier transfers possession of vehicle to manufacturers selling agent. This code is interchangeable with inspection type 5 |
| 99   | Letter of Notification     | Code used to indicate that claim letter has been sent                                                                                      |
| AR   | Arrived in Storage         | Code used for storage yard arrival activity                                                                                                |
| OU   | Removed for Storage        | Code used for storage yard exit activity                                                                                                   |

\* Intermediate inspections 02,07,7R allowed with no destination information.

\*\* Destination inspections 04,4E,4V,4R,08 allowed without origin information.



## JSON Formatted EPOD Delivery Message Example

```
{
  "Delivery":{
    "Carrier":{
      "Code":"CARR1",
      "EquipmentId":"T12345",
      "TrailerNumber":"TRL123",
      "DriverName":"John Johnson"
    },
    "CarrierComment":"Carrier Comment",
    "Comment":"Comment",
    "DeliveryReceipt":{
      "ReferencId":"DOC12345",
      "URL":"https://docs.carrier1.com/receipt/DOC12345"
    },
    "InspectionType":"04",
    "ReferencId":"DLVRY1234",
    "Shipment":{
      "OrderNumber":"679807",
      "DeliveryDateTime":"2013-08-16T10:49:01.038-04:00",
      "DepartureDateTime":"2013-08-16T10:49:01.038-04:00",
      "DestinationCode":"LOC2",
      "OriginCode":"LOC1"
    },
    "SubjectToInspection":true,
    "Vehicle":[
      {
        "VIN":"ABCDEFGHJKM123456"
      },
      {
        "VIN":"ABCDEFGHJKM123457",
        "Damage":[
          {
            "AreaCode":"01",
            "TypeCode":"02",
            "SeverityCode":3,
            "ClassificationCode":"3"
          },
          {
            "AreaCode":"02",
            "Comment":"Optional comment",
            "SeverityCode":4,

```



```
        "TypeCode": "03"
      },
    ],
    "Photo": [
      {
        "ReferenceId": "IMG12345",
        "URL": "https://docs.carrier1.com/photo/IMG12345"
      },
      {
        "ReferenceId": "IMG12346",
        "URL": "https://docs.carrier1.com/photo/IMG12346"
      }
    ]
  }
]
```





## XML Formatted EPOD Delivery Message Example

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<Delivery>
```

```
  <ReferenceId>MSG12345678</ReferenceId>
```

```
  <Carrier>
```

```
    <Code>CARR1</Code>
```

```
    <EquipmentId>T12345</EquipmentId>
```

```
    <TrailerNumber>TRL123</TrailerNumber>
```

```
    <DriverName>John Johnson</DriverName>
```

```
  </Carrier>
```

```
  <Shipment>
```

```
    <OrderNumber>679807</OrderNumber>
```

```
    <OriginCode>LOC1</OriginCode>
```

```
    <DestinationCode>LOC2</DestinationCode>
```

```
    <DepartureDateTime>2012-10-31T14:05:12-05:00</DepartureDateTime>
```

```
    <DeliveryDateTime>2012-11-02T11:55:23-05:00</DeliveryDateTime>
```

```
    <SpecialInstructions></SpecialInstructions>
```

```
  </Shipment>
```

```
  <DeliveryReceipt>
```

```
    <ReferenceId>DOC12345</ReferenceId>
```

```
    <URL>https://docs.carrier1.com/receipt/DOC12345</URL>
```

```
  </DeliveryReceipt>
```

```
  <InspectionType>05</InspectionType>
```

```
  <SubjectToInspection>>false</SubjectToInspection>
```

```
  <Comment>Optional dealer's comment goes here</Comment>
```

```
  <CarrierComment>Optional carrier's comment goes here</CarrierComment>
```

```
  <Vehicle>
```

```
    <VIN>ABCDEFGHJKM123456</VIN>
```

```
  </Vehicle>
```

```
  <Vehicle>
```

```
    <VIN>ABCDEFGHJKM123457</VIN>
```

```
    <Damage>
```



```
<AreaCode>01</AreaCode>
<TypeCode>02</TypeCode>
<SeverityCode>3</SeverityCode>
<ClassificationCode>3</ClassificationCode>
</Damage>
<Damage>
  <AreaCode>02</AreaCode>
  <TypeCode>03</TypeCode>
  <SeverityCode>4</SeverityCode>
  <Comment>Optional damage comment goes here</Comment>
</Damage>
<Photo>
  <ReferenceId>IMG12345</ReferenceId>
  <URL>https://docs.carrier1.com/photo/IMG12345</URL>
</Photo>
<Photo>
  <ReferenceId>IMG12346</ReferenceId>
  <URL>https://docs.carrier1.com/photo/IMG12346</URL>
</Photo>
<Photo>
  <ReferenceId>IMG12347</ReferenceId>
  <URL>https://docs.carrier1.com/photo/IMG12347</URL>
</Photo>
</Vehicle>

</Delivery>
```